Hochschule für Wirtschaft und Recht Berlin
Berlin School of Economics and Law

Product Discovery with Design Thinking

Wi-Inf-M09-F02 Full-Stack Web Development

Prof. Dr. Alexander Eck

Semester plan (subject to change)

#	Week	Project Work	Lecture (Thu)	Lab (Fri)
1	06.04. - 10.04.	--	Kick-off	Kick-off (short), tools setup
2	13.04. - 17.04.	Get up to speed with Python	Git + GitHub workflows	Repository setup, incl. GitHub Pages
3	20.04. - 24.04.	Exercise collaborative workflows	Full-stack web development with Flask	Python Q&A + implement example app
4	27.04. - 01.05.	Ideate potential problem spaces	Product Discovery / Design Thinking	No Lab
5	04.05. - 08.05.	**Assignment** (05.05. 23h)	Design Decisions + Markdown syntax	Assignment peer reviews
--	11.05. - 15.05.	Product Discovery	No Lecture	No Lab
6	18.05. - 22.05.	Product Discovery **Assignment re-work** (19.05. 23h)	Flask URL path routing (based on HTTP)	App scope peer reviews
7	25.05. - 29.05.	Implement most basic feature	No Lecture	App skeleton (at least: all routes, incl. API)
8	01.06. - 05.06.	Product Delivery + Design Decisions	User Interfaces with WTForms + Bootstrap	Happy Path (e.g., as screen-flow)
9	08.06. - 12.06.	Product Delivery + Design Decisions	Relational databases with SQLAlchemy	Data Model (DB schema needed for app)
10	15.06. - 19.06.	Product Delivery + Design Decisions	First Submission Q&A (15 min / team)	First Submission Q&A (15 min / team)
11	22.06. - 26.06.	**First Submission** (23.06. 23h)	Selected topics in web development	Peer-conducted exam preparation
12	29.06. - 03.07.	*Contingency*	Selected topics in web development	Peer-conducted exam prep + reflection
13	06.07. - 10.07.	Prepare for examination	**Oral Examination** (09.07.)	**Oral Examination** (10.07.)
14	13.07. - 17.07.	Finishing touches	**Oral Examination** (16.07.)	**Oral Examination** (17.07.)
--	20.07. - 24.07.	**Final Submission** (21.07. 23h)		

At the end of this session, you will be able to…

Recall the phases of Product Discovery and Delivery, and the four Product Dimensions

Apply basic Design Thinking tools to ideate your web-based application

Recall the Design Thinking micro process and the purpose of each activity

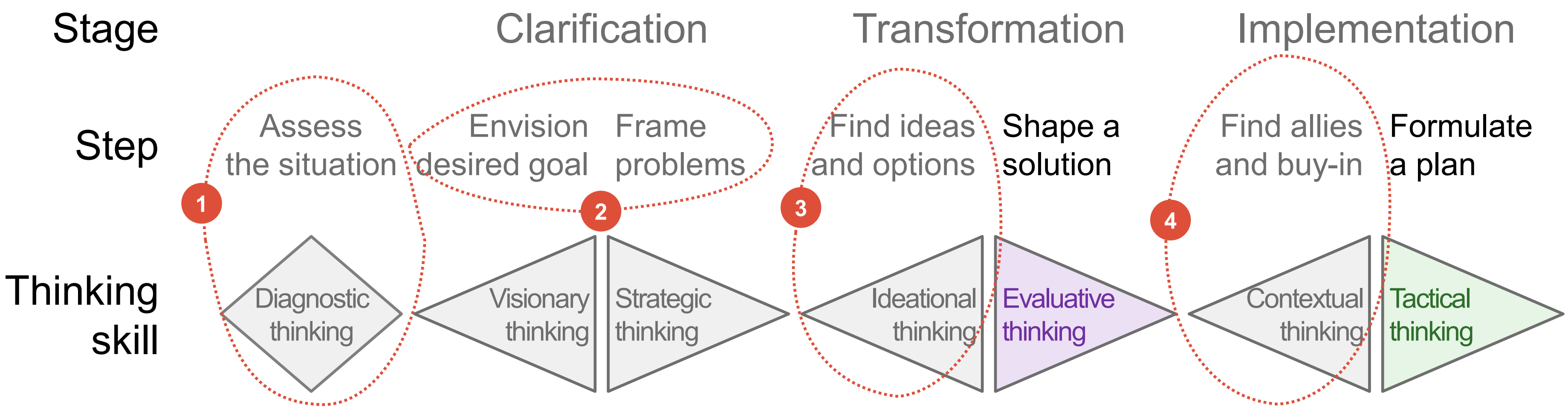
Start working on your web-based application with a solid value proposition in mind

Creative Problem Solving: identify the right problem and find convincing solutions = <u>Product Discovery</u>

Stage		Clarification		Transformation		Implementation	
Step	Assess the situation[2]	Envision desired goal	Frame problems	Find ideas and options	Shape a solution	Find allies and buy-in	Formulate a plan
Thinking skill[1]	Diagnostic thinking	Visionary thinking	Strategic thinking	Ideational thinking	Evaluative thinking	Contextual thinking	Tactical thinking
Affective skill[3]	Mindfulness	Dreaming	Sensing gaps	Playfulness	Avoid premature closure	Sensitivity to environment	Tolerance for risk

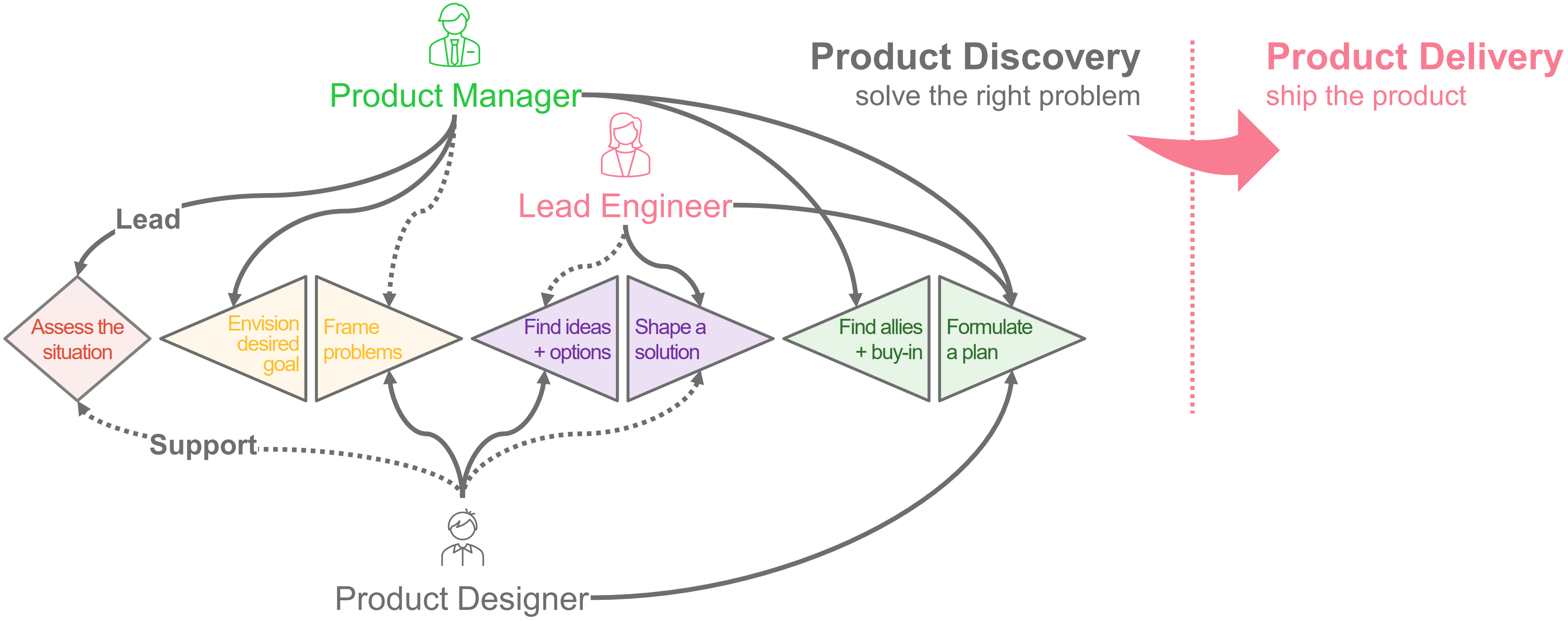
1. Refers to mode of thinking | 2. Often second step after “envision desired goal” | 3. Refers to state of mind
Adapted from: Puccio, Mance & Murdock (2011). Creative leadership (2nd ed.). SAGE Publications.

Many organizations skip most steps: this leads to solving the wrong problems with unconvincing or non-viable solutions



1. "I know my business inside-out": risk to lose touch with reality; risk to overlook key constraints
2. "I already know the problem": risk to set inconsistent goals over time; risk to address symptoms, not root-cause
3. "We don't need wild ideas over here": risk to produce incremental rather than breakthrough innovations
4. "I expect people to execute on decisions": risk to fail due to overlooked barriers; risk of mis-understanding

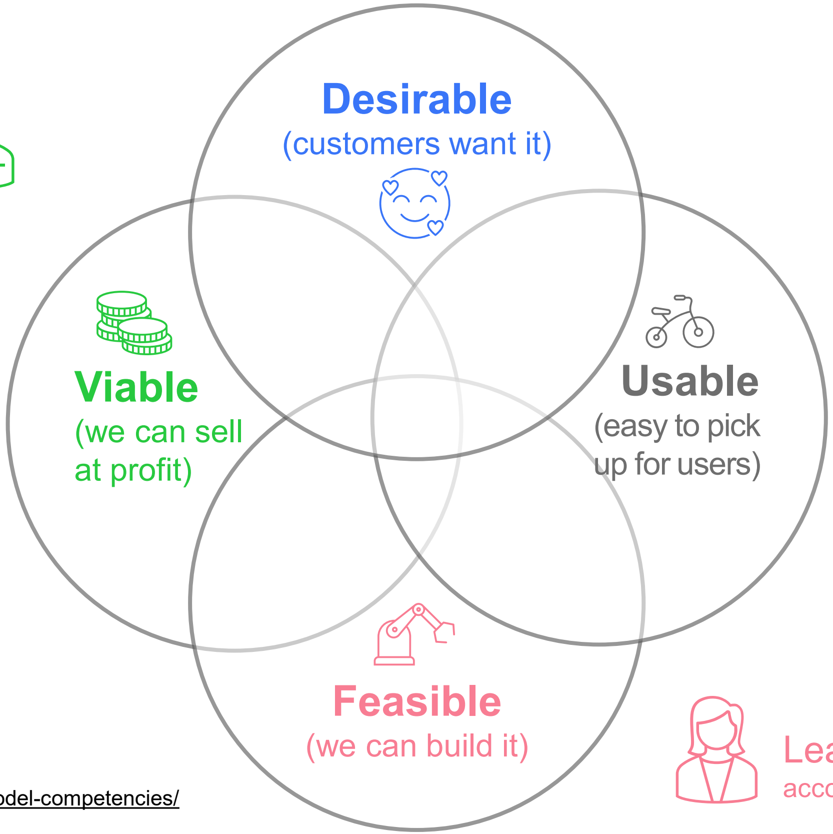
In Product Discovery, the skills mix of the Product Team helps to execute every step of Creative Problem Solving



Note: “Lead” and “Support” are indicative: in effective teams, all three roles contribute at every step

Product Team with 3 key profiles owns Product Discovery (solve problem) + Product Delivery (ship product)

Product Manager
accountable for outcome

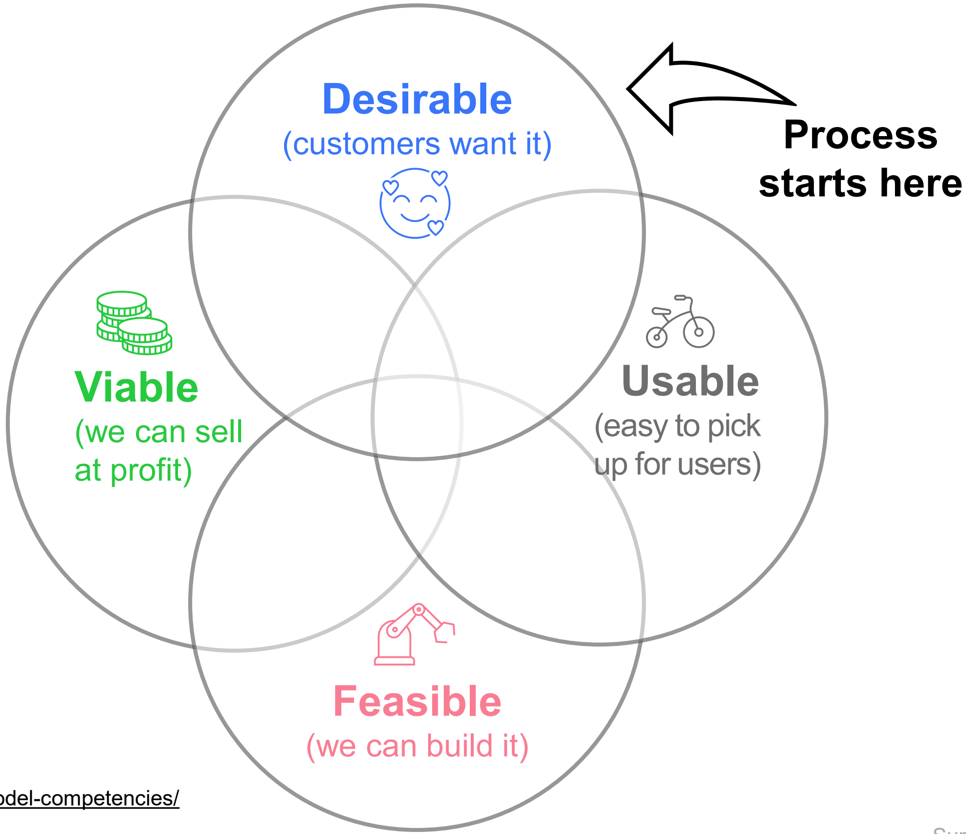


Product Designer
accountable for experience

Lead Engineer
accountable for delivery

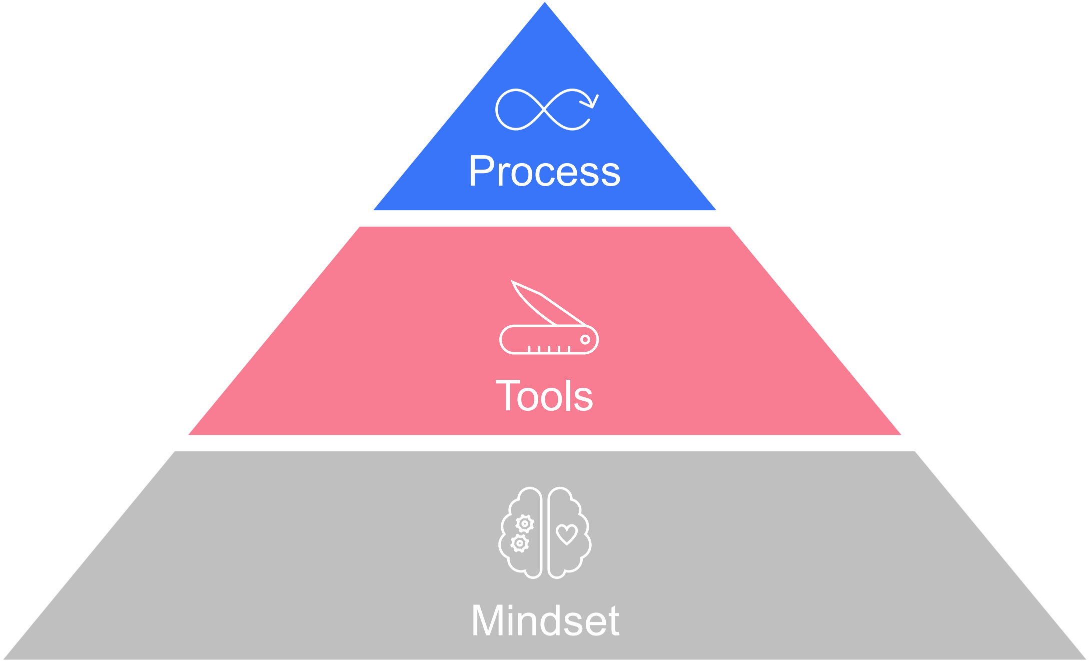
Adapted from https://www.svpg.com/product-model-competencies/

Design Thinking integrates human needs, technological possibilities and business requirements



Adapted from https://www.svpg.com/product-model-competencies/

Design Thinking (DT) is an innovation method with several perspectives

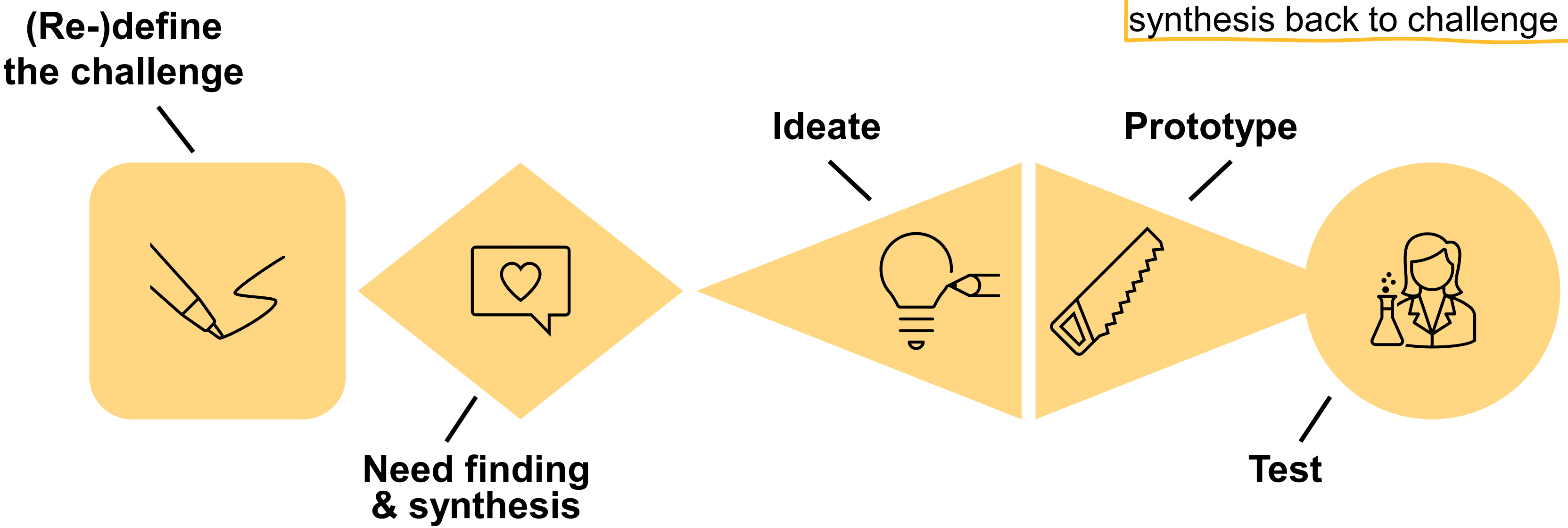


Design Thinking was conceived by engineers for engineers

- Design Thinking has its roots at the Department of Mechanical Engineering at Stanford University
- It was conceived in the 1960s to teach the “why” and “for whom”, in addition to the traditional engineering curriculum of “what” and “how”
- The 3-terms / 1-year long course “ME310” has been taught since 1967
- After many rounds of refinement, DT has become a human-centered method that also integrates technology and business aspects

During a project, the design team will iterate through this "micro process" multiple times

You can (and will) go back and forth between these steps, e.g., from synthesis back to challenge definition



Note: Multiple Design Thinking process models exist, but they are all very similar. This is my own visualization, which is inspired by: Uebernickel et al. 2015. Design Thinking.

Show me your wallet and I tell you who you are (point of view template)

User

Alexander, an avid biker

Need

Means to get quick access to his important cards (credit, ID, ...) in a small form-factor that is not easily lost

Insight

Alexander doesn't use cash anymore, since he switched to card-only pay. Also, he tries to minimize size + weight of the items he carries with him.

Note: This template can help formulate a good "point of view". It is okay not to use it. (This is true for all templates that I am going to present.)

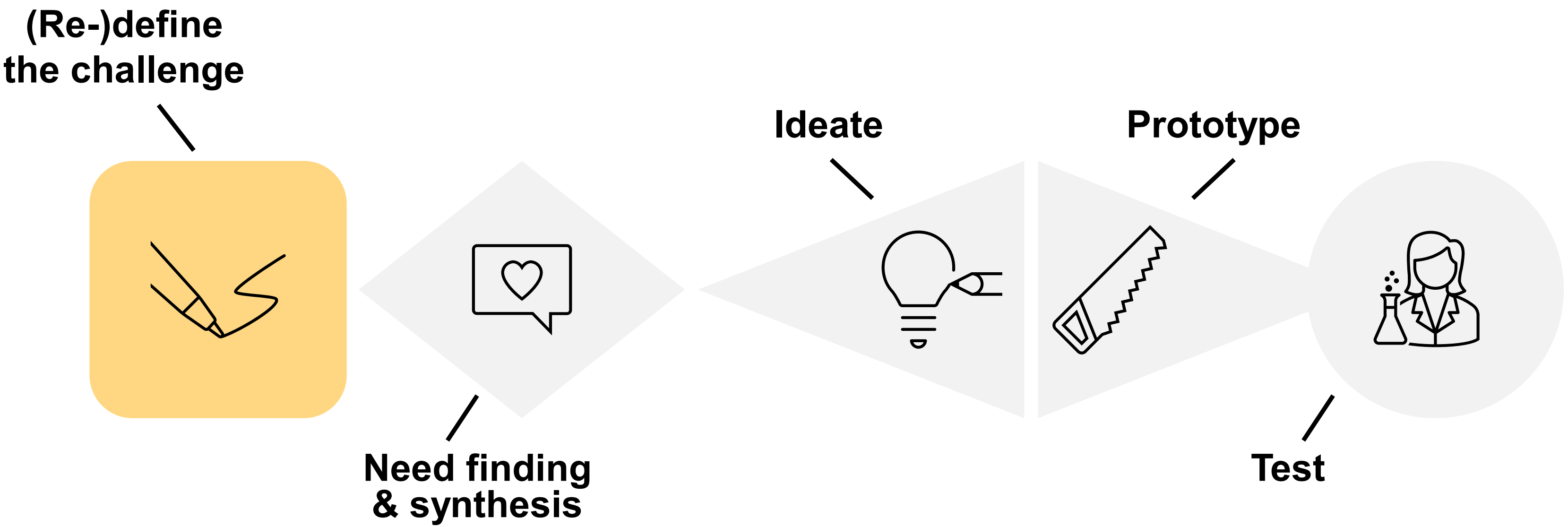
Show me your wallet and I tell you who you are

10 min
time budget

- Come together in groups of 4 (or 5)
- Select one person to be interviewed & observed
- First, develop a “point of view” for this person which is related to their wallet
  - **3 min** interview
  - **2 min** synthesis: write down the “point of view”

- Next, design a solution that addresses the need and test it with your target user
  - **4 min** ideation & prototyping
  - **1 min** testing

(Re-)define the challenge



Write down and agree on a design challenge without hinting at or including a solution approach

How might we help

freelancers (target user group)

to

keep track of their task allocation (need to be addressed)

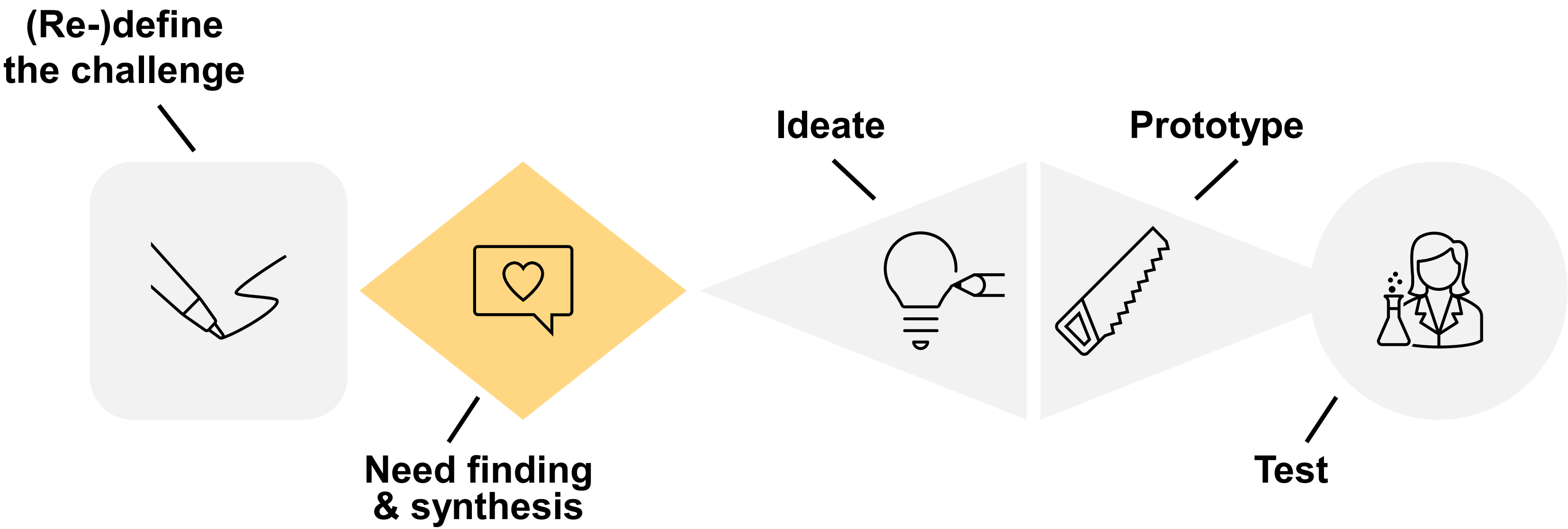
considering [optional]

a tendency to procrastinate and their need to prioritize tasks

(factors that make the challenge even more challenging)

- As you iterate through the micro-process, you are encouraged to change the design challenge, e.g.:
  - **Pivot:** you find out that the root problem is different, or that there is a more pressing human need to be addressed.
  - **Refine:** as you learn more about the problem / solution domain, you typically want to break up the design challenge into multiple sub-challenges.

Need finding & synthesis



Needs are verbs, solutions are nouns: needs last longer than any specific solution

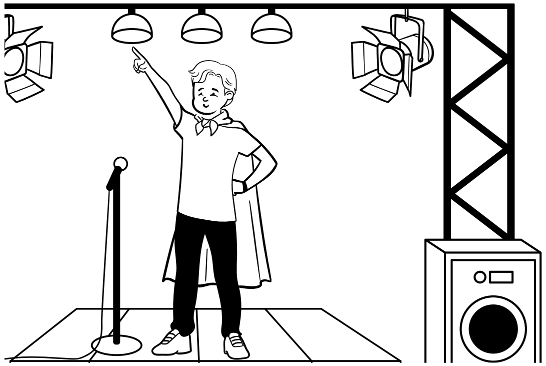
Source: https://www.chinadaily.com.cn/business/2017-09/21/content_32290413_9.htm

To discover needs, engage and empathize with users – there is no value creation without empathy

Ask

Observe

Engage



+ Desk research, competitor analysis and other "secondary research" methods are also great to identify needs of your target users

What to ask about and observe: the AEIOU framework

A **Activities:** Which problem are people solving, what might frustrate them?

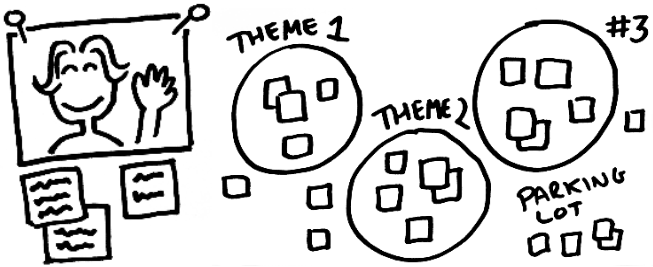
E **Environment:** What is the character and function of the focal physical / virtual space?

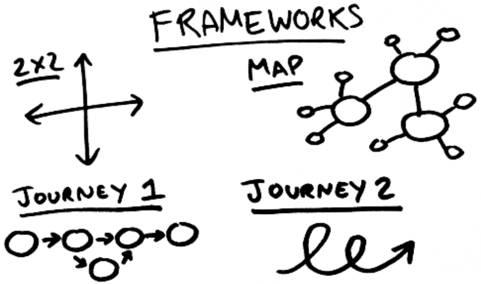
I **Interactions:** What is happening between elements, i.e., people and objects?

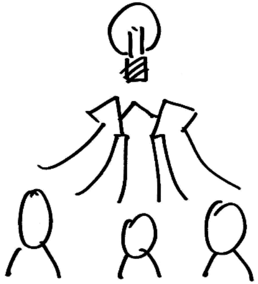
O **Objects:** Which objects are (available for) use(d), what are their characteristics and conditions?

U **Users:** Who is there? What are their position in space, characteristics, roles?

How to synthesize research material from need finding





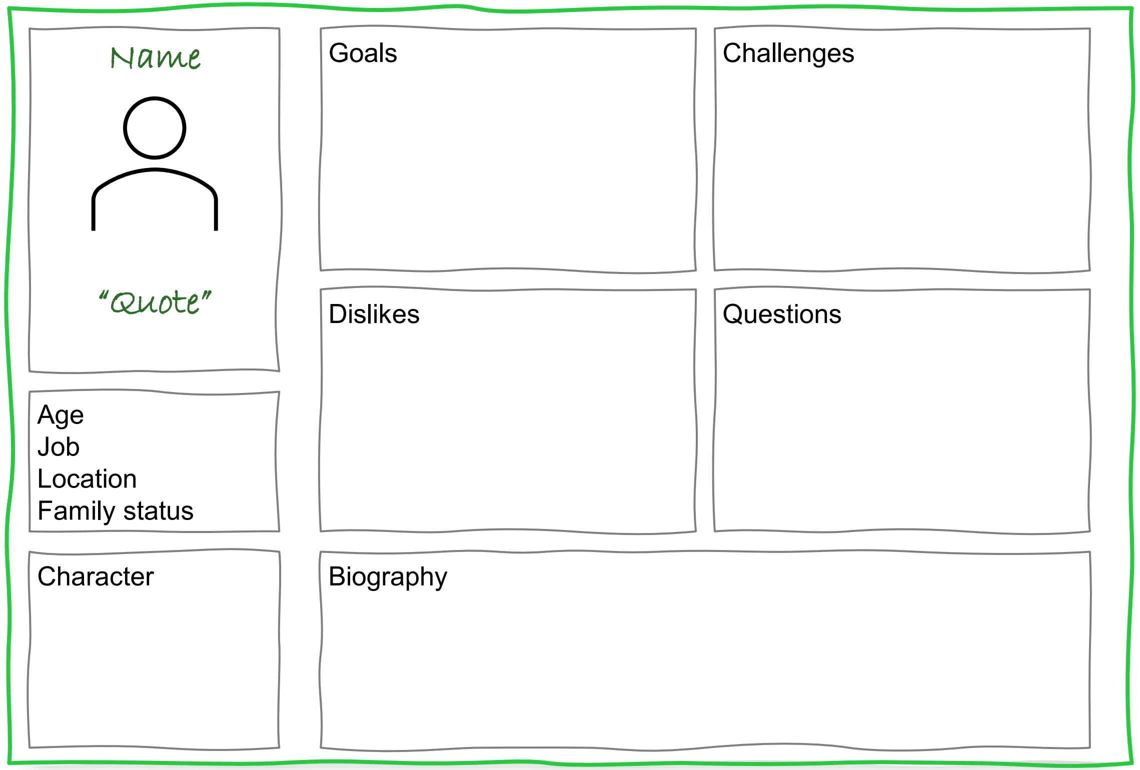


Get immersed in the research material: tell stories, find patterns, look for insights

Find good visualizations to structure and refine your learnings

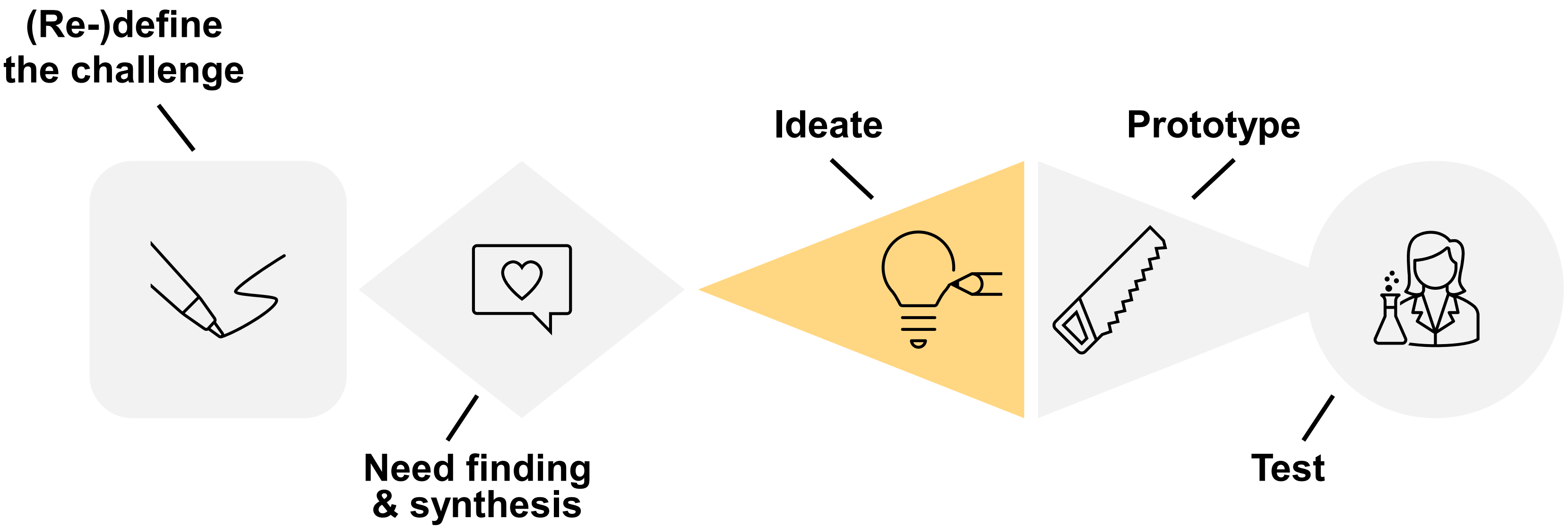
Refine your design challenge as needed

Persona: representative profile of traits, behaviors, beliefs, … of your target user group – captures needs

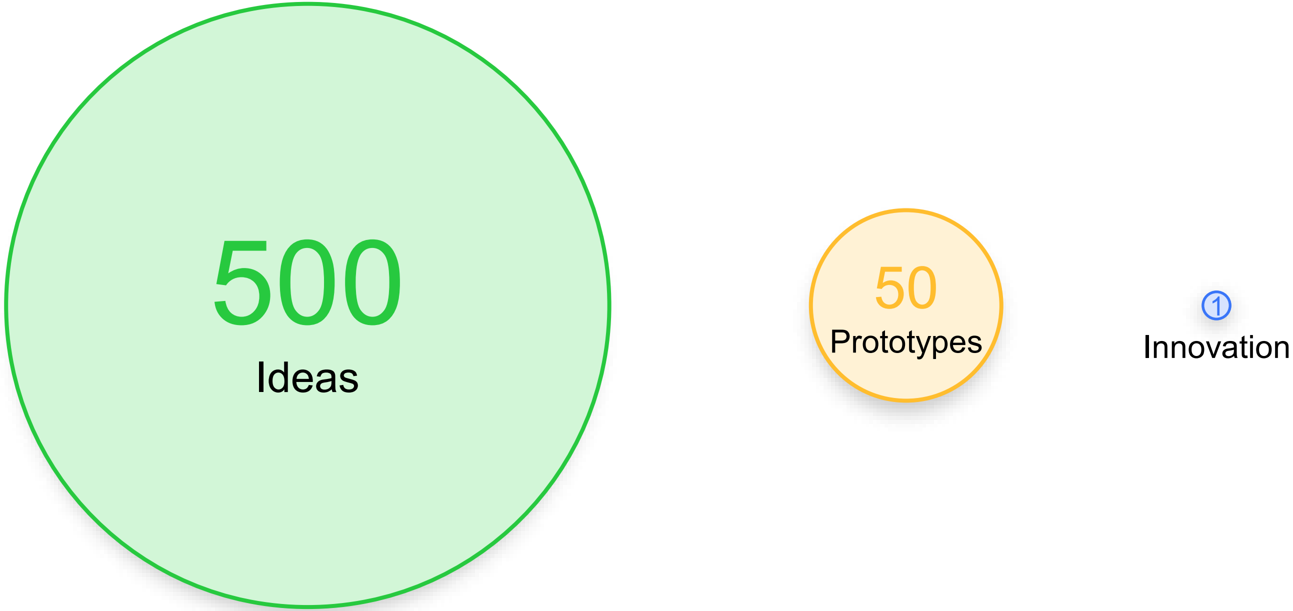


- Personas "speak" about common characteristics of your target user group(s): they are not made up, but represent real people
- They help the design team and stakeholders to create a shared understanding of the user
- Personas are useful in guiding decisions about product features, visual design, …
- They also help targeting your marketing activities during go-live
- The categories on the left are generic examples: you will want to design them according to your specific design challenge

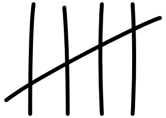
Ideate


(Re-)define
the challenge
Need finding
& synthesis
Ideate
Prototype
Test

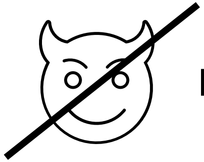
“If you want to have good ideas, you must have many ideas” – Linus Pauling, receiver of two Nobel Prizes

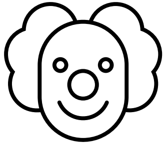


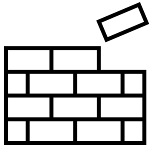
Brainstorming is a simple and popular method to generate many ideas quickly – a few tips to keep in mind

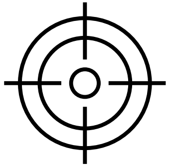
 Go for quantity

 Be visual

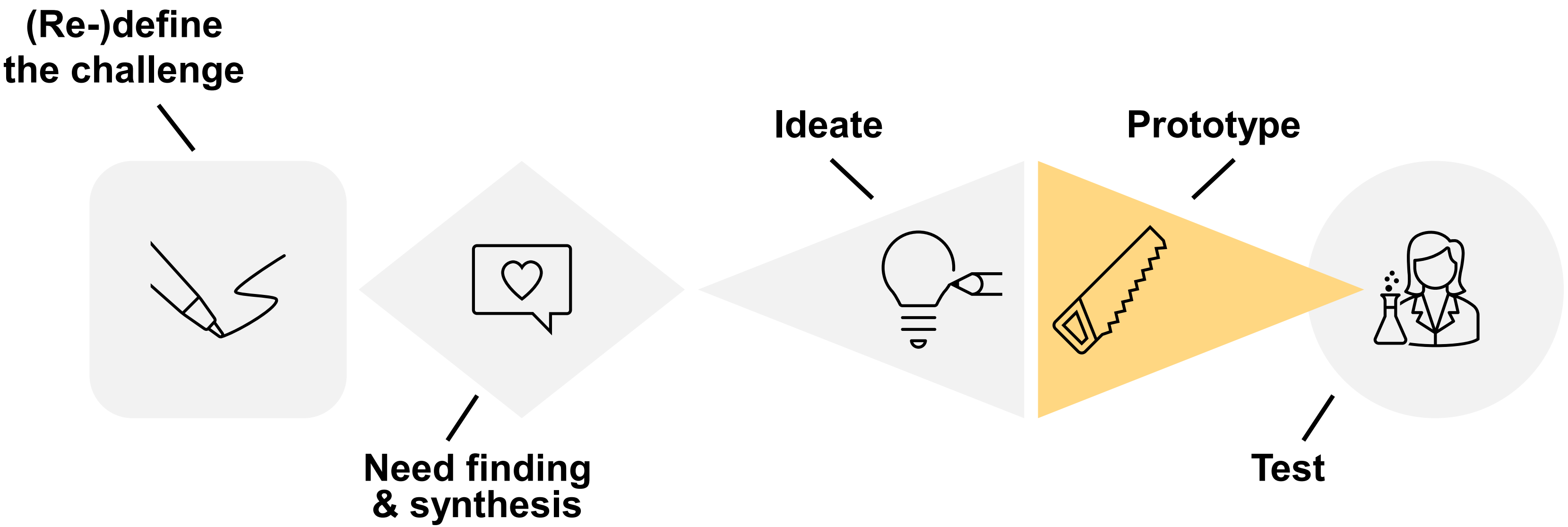
 Defer judgement

 Encourage wild ideas

 Build on ideas of others

 Stay focused on the topic

Prototype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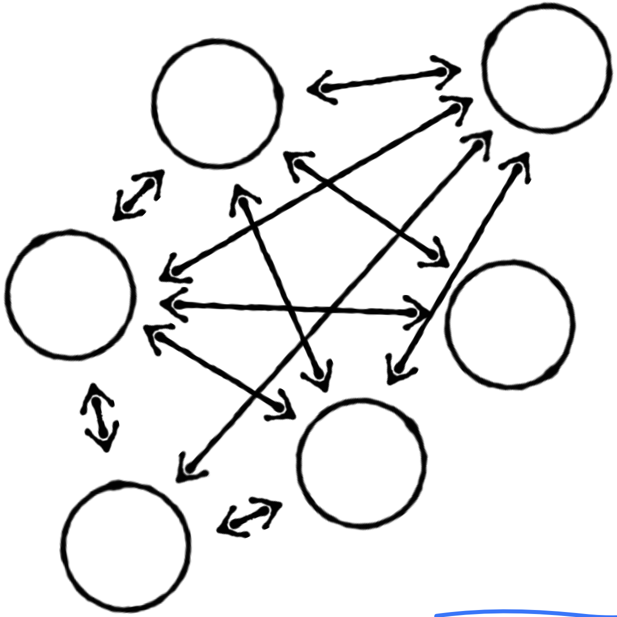
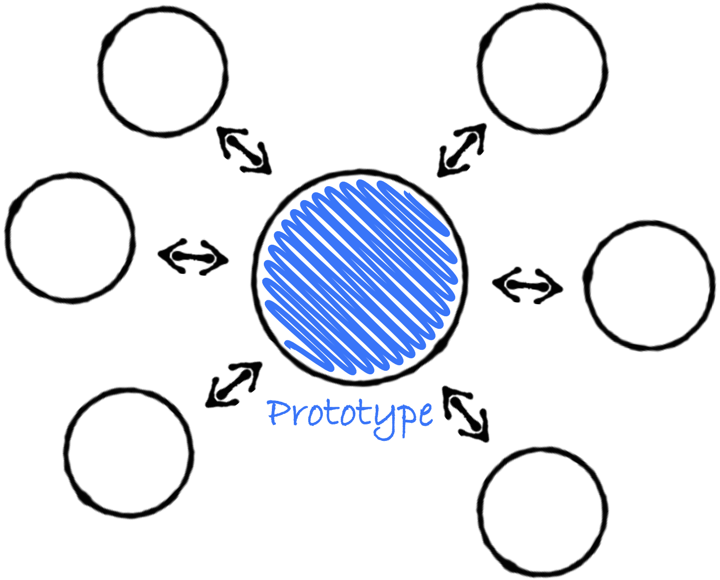
A prototype is anything that helps you make an idea tangible, unify discussions and get feedback



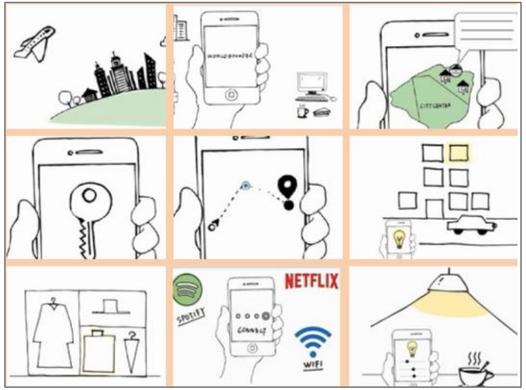


Prototyping is the key instrument to drive progress in a Design Thinking project!

A prototype is anything that helps you make an idea tangible, unify discussions and get feedback

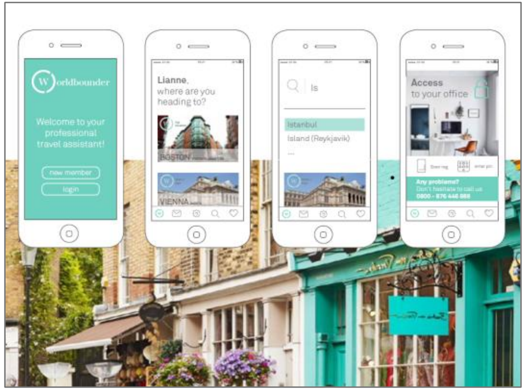
A few prototype examples:

Comic

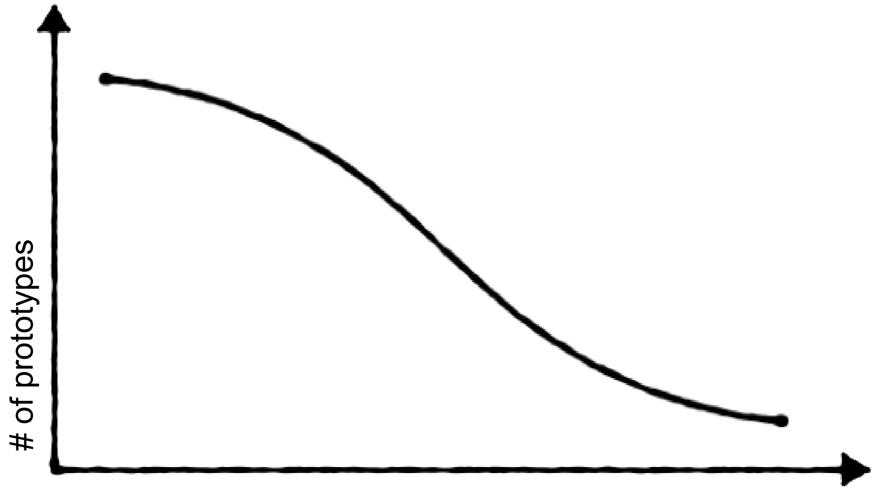


Paper prototype

Click dummy



To test early & often, you will create many low-resolution prototypes, then converge to high-resolution prototypes



Low-resolution prototypes

Role-play
Scribble
Comic
Paper prototype
…

High-resolution prototypes

Prototype combination
Customer journey
Product video
Click dummy
…

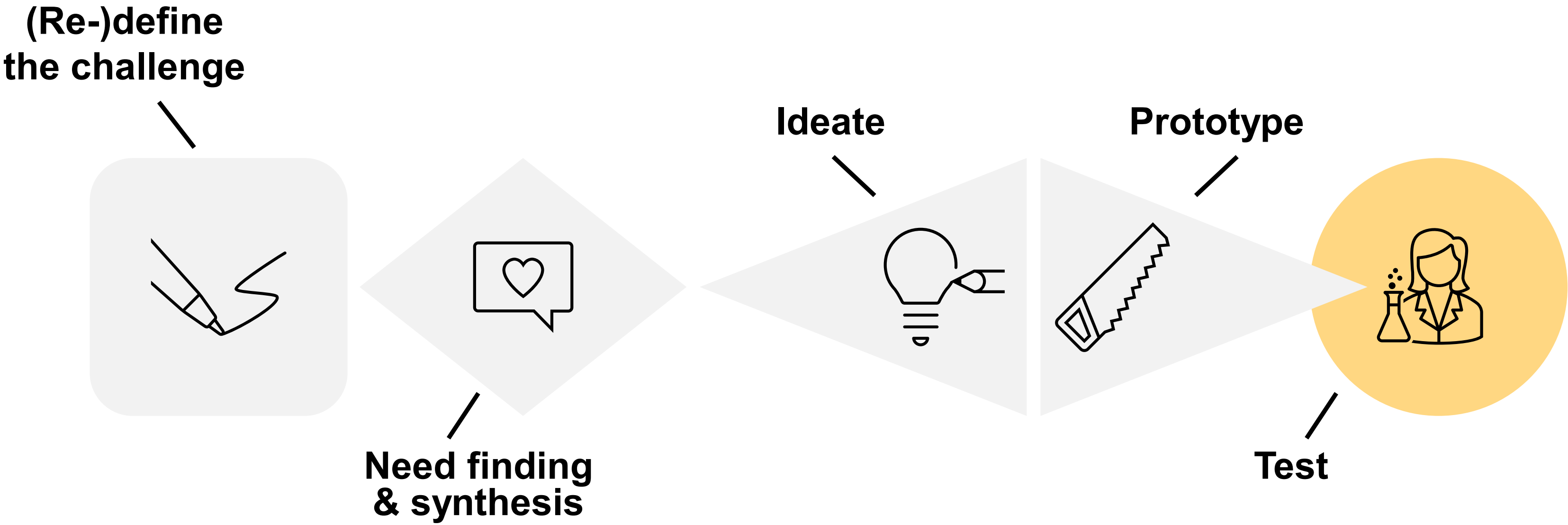
Low-resolution prototypes

- Verify a specific idea / critical hypothesis fast (think minutes, not months)
- Low emotional attachment of the design team, since they are created quickly/ cheaply and typically not aesthetically pleasing

High-resolution prototypes

- Combine the most successful prototypes / validated ideas to more complex artifacts
- Users may think this is already the finished product
- Usually, your minimum viable product (MVP) will launch with fewer features than your most sophisticated high-resolution prototype

Test


(Re-)define the challenge
Ideate
Prototype
Need finding & synthesis
Test

Testing uses similar tools to need finding, but now you introduce an artifact you designed; testing purposes:

Evaluation

Evaluate the prototype for specific qualities, e.g., usability, usefulness, durability, weight, … - often by the design team

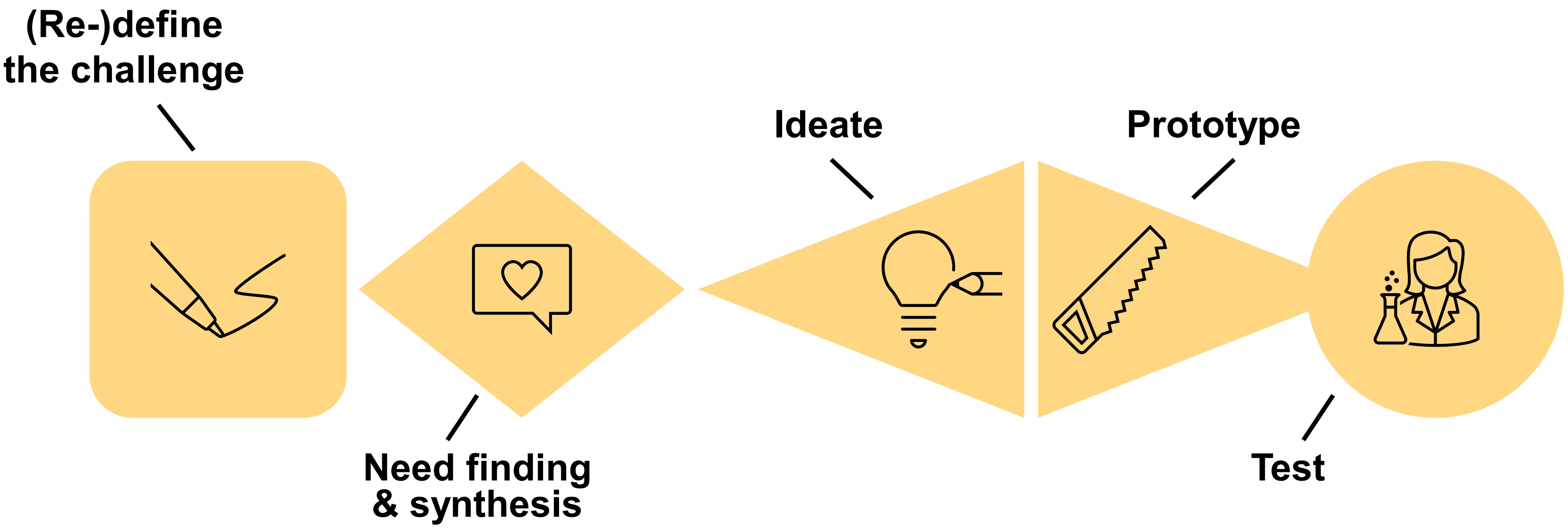
Inspiration

Inspire the team and users to propose new ideas, suggest improvements, or find out hidden needs

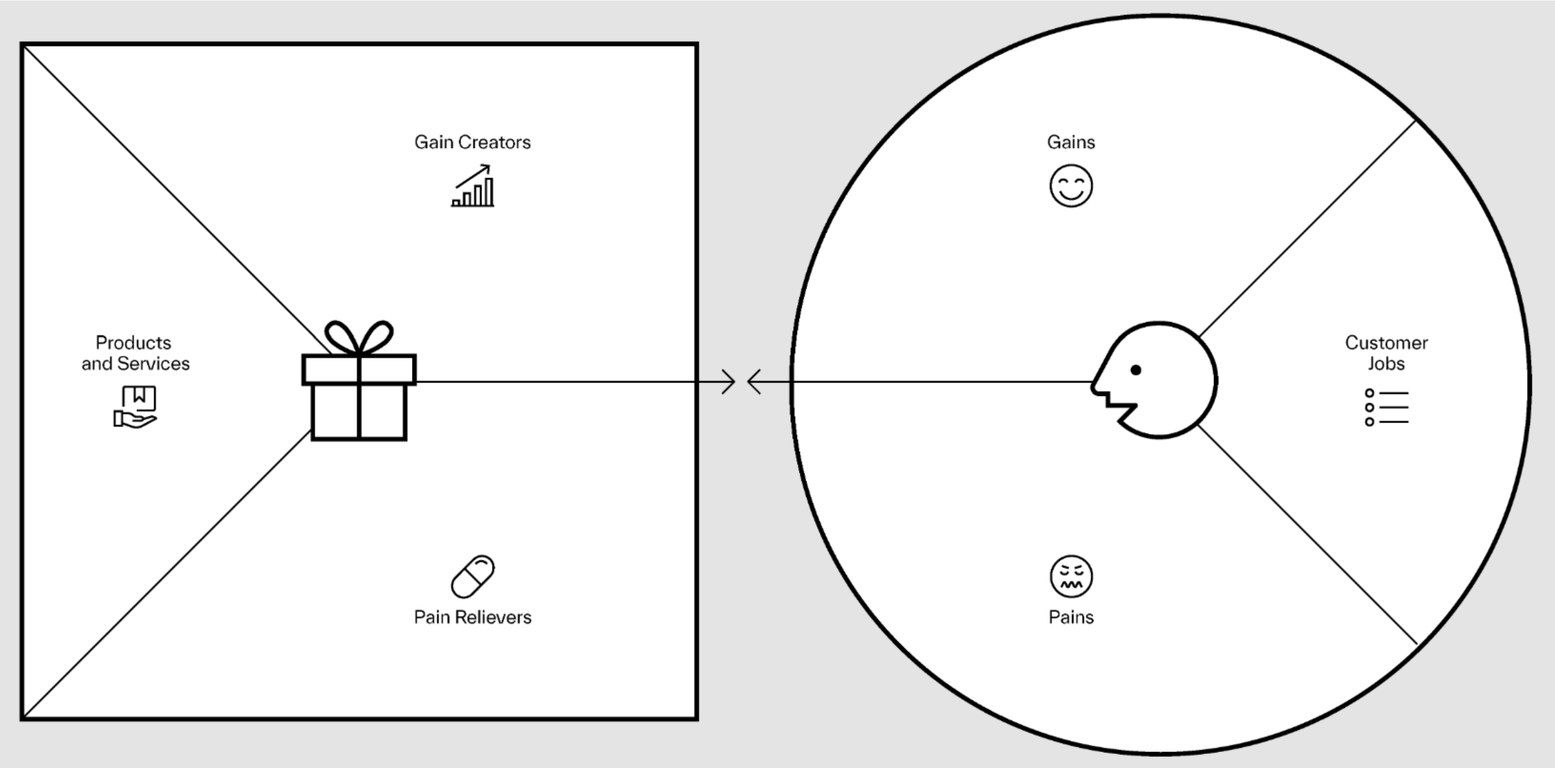
Validation

Validate with the target user group to which extent the expected value is really created, or what is missing

Recap: the Design Thinking “micro process”



Bonus: Value Proposition Canvas helps consolidate and create insights throughout the “micro process”, esp. in early phases



Source: https://www.strategyzer.com/library/the-value-proposition-canvas

Value Proposition Canvas: match customer needs with offering

Customer Segment

Which customers do we address?

Match

Value Proposition

What do we offer these customers?

To-Dos

What are our customers trying to do or achieve?

Gains

How do they measure success, what are desired outcomes?

Pains

What frustrations, risks, or obstacles do they face?

Features

What are we offering?

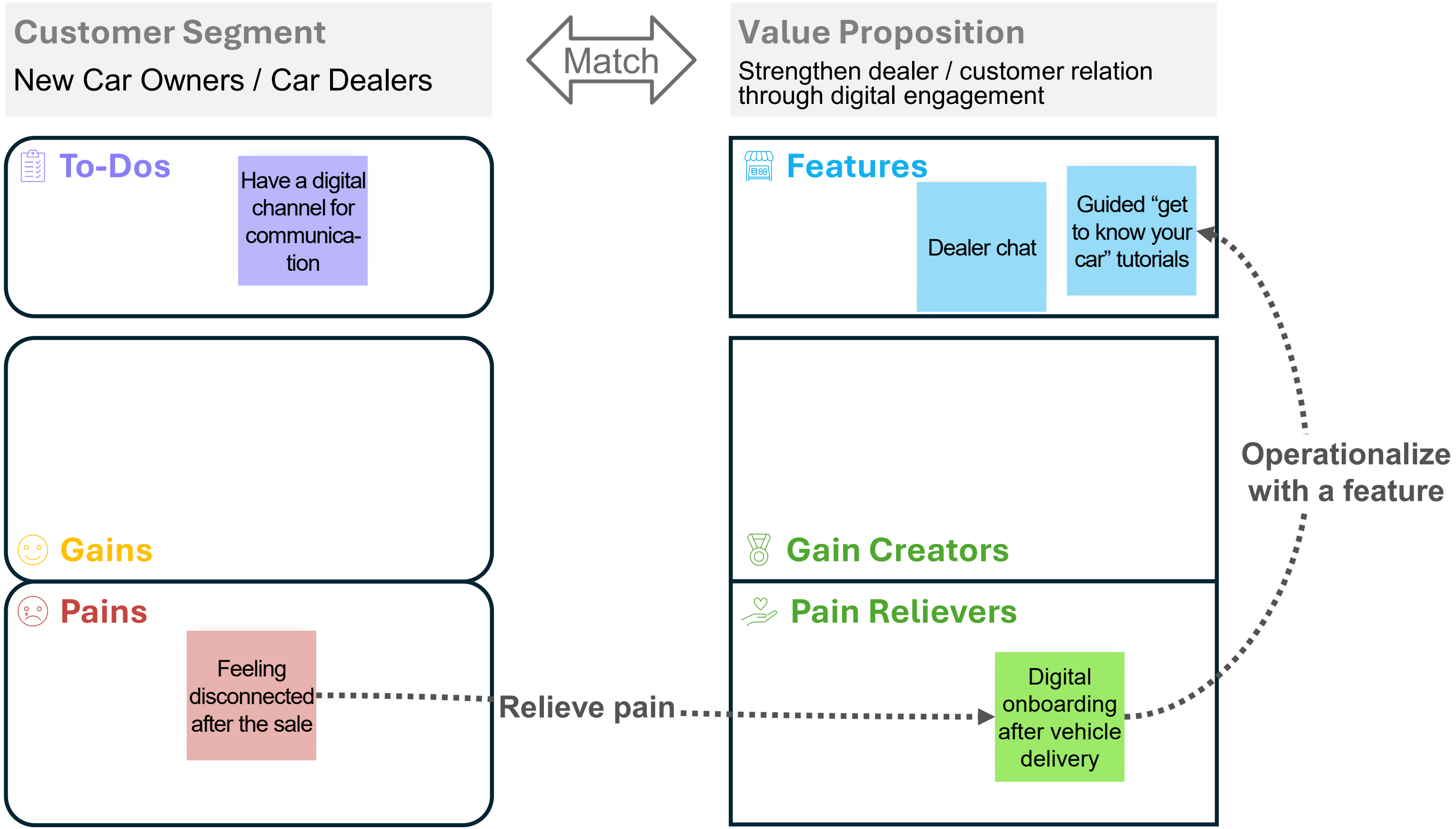
Gain Creators

How does our solution help customers be successful?

Pain Relievers

How does our solution remove customer pains?

Exemplary flow



Thank you!

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